

Google Cloud Platform (GCP)

1



Google Cloud

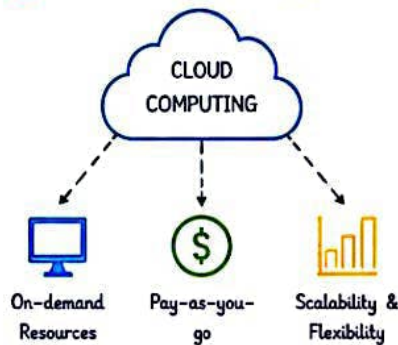
Google Cloud Platform is a suite of cloud computing services that runs on the same infrastructure that Google uses for its own end-user products.

1 What is Google Cloud?



- Google Cloud Platform (GCP) offers a broad set of cloud services including computing, storage, data analytics, machine learning, and more.
- It helps businesses build, deploy, and scale applications securely and efficiently.
- GCP runs on Google's premium global network infrastructure.

2 Cloud Computing



- Deliver IT resources over the internet.
- On-demand self-service.
- Pay only for what you use.
- Scale up or down in minutes.
- High availability and reliability.

3 GCP Benefits



Secure by design with advanced security.



High performance and reliable infrastructure.



Cost-effective with sustained use discounts.



Global scale with low latency network.



Innovative services in AI/ML, Data, and more.

4 Global Infrastructure



40+ Regions across the world



100+ Zones for high availability



Global network built for speed



Edge locations for low latency



GCP's global infrastructure ensures your applications are fast, reliable, and always available to your users.

5 Popular GCP Services



Compute Engine



Cloud Storage



Cloud SQL



BigQuery (Data Warehouse)



Google Kubernetes Engine (GKE)



Cloud Functions



Cloud Run



Vertex AI (ML Platform)



Cloud IAM



Cloud Monitoring



Cloud Firestore



Cloud Pub/Sub



REMEMBER

- GCP is built on Google's global infrastructure.
- Pay-as-you-go - no upfront cost.
- You manage security in the cloud.
- Choose the right service for the right workload.
- Always design for high availability and scalability.
- Use monitoring and logging for proactive management.



BEST PRACTICES

- ✓ Use Least Privilege with IAM roles.
- ✓ Enable Cloud Monitoring & Logging.
- ✓ Use managed services wherever possible.
- ✓ Follow Well-Architected Framework.
- ✓ Enable Billing Alerts and Budgets.
- ✓ Keep resources in the right region.
- ✓ Regularly review and clean up unused resources.



GCP Global Infrastructure

2



Google Cloud

Google Cloud operates one of the world's largest, fastest and most reliable global networks.

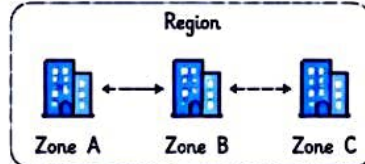
It helps you build scalable, high-performance applications with low latency and high availability.

1 Regions



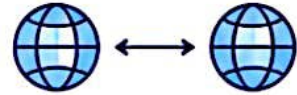
- A Region is a specific geographic location.
- Each region contains one or more zones.
- GCP has 40+ regions across the world.

2 Zones



- Zones are isolated locations within a region.
- Designed to protect applications from data center failures.
- Use multiple zones for high availability.

3 Multi-Region



- Multi-Region provides high availability across multiple regions.
- Data is automatically replicated across regions.
- Used for services like Cloud Storage, Cloud Spanner, BigQuery.

4 Edge POPs



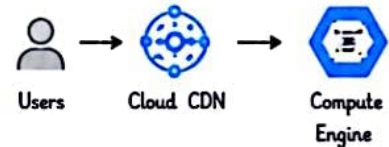
- Google has 200+ Edge Points of Presence (POPs).
- Delivers content closer to users for lower latency.
- Improves performance and reliability.

5 Cloud CDN



- Content is cached at Google's edge locations.
- Reduces latency and origin load.
- Integrates with HTTPS Load Balancer, Cloud Storage and Cloud CDN.

6 Real-world Example



- A global e-commerce app uses Cloud CDN to deliver images and videos.
- Static content served from edge POPs.
- Dynamic content served from nearest region.

GCP Global Footprint



Key Highlights

- ✓ 40+ Regions
- ✓ 100+ Zones
- ✓ 200+ Edge POPs
- ✓ 140+ Network Edges
- ✓ Fast fiber network backbone

REMEMBER

- ✓ A Region contains one or more Zones.
- ✓ Deploy across multiple zones for high availability.
- ✓ Use Multi-Region for global availability of data.
- ✓ Edge POPs reduce latency and improve user experience.
- ✓ Cloud CDN caches content at edge for speed.
- ✓ Choose the right location for compliance and performance.



BEST PRACTICES

- ✓ Design for high availability across multiple zones.
- ✓ Store data in the nearest region for users.
- ✓ Use Cloud CDN for static content delivery.
- ✓ Enable Multi-Region for critical data and services.
- ✓ Monitor latency and performance using Cloud Monitoring.
- ✓ Plan for DR using Multi-Region and backups.



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Google Cloud IAM

3

Identity and Access Management (IAM) in GCP helps you securely manage who can access which resources and what actions they can perform.

1 Cloud IAM



- Centralized system to manage access to GCP resources.
- Based on identities (who) and permissions (what).
- Applies at Organization, Folder, Project and Resource level.

2 Users



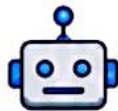
- Individuals who access GCP using their Google account.
- Grant access using IAM roles.
- Can be members of groups.

3 Groups



- Collection of users.
- Assign IAM roles to groups instead of individual users.
- Simplifies access management at scale.

4 Service Accounts



- Special identities for applications and workloads.
- Used by VMs, Apps, Functions, APIs, etc.
- Authenticate using keys or Workload Identity.

5 IAM Roles



- Collection of permissions.
- Predefined roles (e.g., Viewer, Editor, Owner).
- Custom roles can be created as per requirement.

6 Organization Policy



- Set guardrails and constraints for your organization.
- Enforce policies like allowed services, regions, VM types.
- Helps maintain compliance and governance.

7 Workload Identity



- Securely access Google Cloud APIs without service account keys.
- Maps Kubernetes Service Accounts to GCP Service Accounts.
- Used with GKE, Cloud Run, etc.

8 IAM Conditions



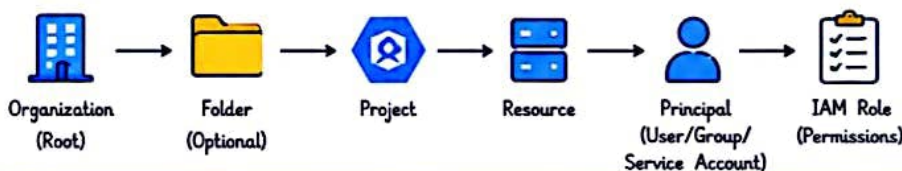
- Add conditions to IAM bindings.
- Control access based on attributes like IP, time, resource, etc.
- Example: Allow access only from corporate IP range.

9 Least Privilege



- Grant only the permissions required to perform a task.
- Reduces risk of unauthorized access and security incidents.
- Review access regularly.

GCP IAM Access Hierarchy



Key Concepts

- ✓ Who → Principal
- ✓ What → Role (Permissions)
- ✓ Where → Resource
- ✓ How → Binding
- ✓ Why → Access

REMEMBER

- IAM is resource-centric.
- Access is inherited from parent to child.
- Use groups to manage access efficiently.
- Avoid using Service Account Keys. Use Workload Identity.
- Regularly audit IAM policies and permissions.



BEST PRACTICES

- ✓ Follow the principle of Least Privilege.
- ✓ Use predefined roles whenever possible.
- ✓ Use IAM Conditions for fine-grained access.
- ✓ Enable Organization Policies for governance.
- ✓ Use Workload Identity for applications on GKE/Cloud Run.
- ✓ Review and remove unused accounts and roles.
- ✓ Enable audit logs for IAM policy changes.



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Compute Engine (Virtual Machines)

4

Compute Engine lets you run virtual machines (VMs) on Google's infrastructure. VMs are highly configurable, scalable and available.

1 VM Basics



- Virtual machines running in Google's data centers.
- Fully configurable compute resources.
- Choose OS (Linux/Windows) and install applications.
- Pay for what you use.

2 Machine Types

- Predefined machine configurations.
- Optimize for general purpose, compute optimized, memory optimized, GPU, etc.
- Custom machine types also available.

Examples

- E2 General purpose
- N2 Balanced
- C3 Compute optimized
- M2 Memory optimized
- A2 AMD based
- A3 GPU optimized

3 Persistent Disk



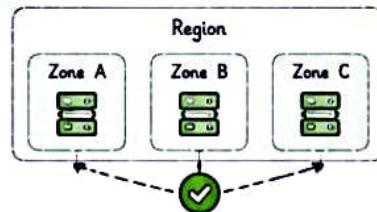
- Block storage for your VMs.
- Data is durable even if VM is deleted.
- Types: Standard, Balanced, SSD, Extreme SSD.
- Can resize disk without downtime.
- Automatically replicated within a zone.

4 Sole-Tenant Node



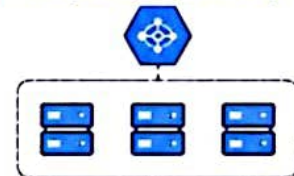
- Your VMs run on a dedicated physical server.
- No multi-tenancy – highest performance and isolation.
- Useful for licensing, compliance and regulatory needs.

5 Availability



- Deploy across multiple zones for high availability.
- If one zone fails, VMs in other zones keep running.
- Use regional resources like Persistent Disk, Load Balancer, Cloud SQL for HA.

6 Managed Instance Group (MIG)



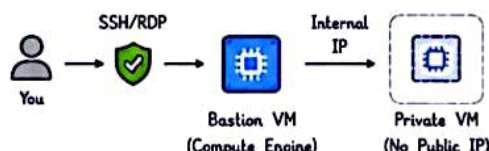
- Automatically manage VM instances.
- Auto-healing, auto-scaling, rolling updates.
- Distribute across zones for availability.
- Works with Load Balancer.

7 Startup Scripts



- Scripts run when a VM starts.
- Automate installation, configuration and updates.
- Use metadata to pass scripts.
- Supports Linux (bash) and Windows (PowerShell).

8 Bastion Host



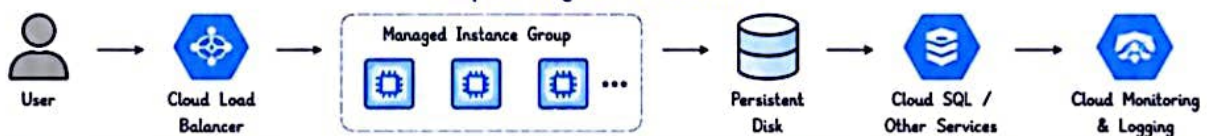
- Secure way to access VMs in a private network.
- Bastion VM has a public IP; internal VMs do not.
- Improves security and reduces attack surface.

9 Snapshots



- Point-in-time copy of a disk.
- Used for backup and recovery.
- Incremental – saves only changed blocks.
- Can create disks from snapshots.

Compute Engine Architecture



REMEMBER

- Compute Engine provides IaaS – full control of your VMs.
- Use multiple zones for high availability.
- Persistent Disk is the primary storage for VMs.
- Use MIG for auto-scaling and auto-healing.
- Use Bastion Host to securely access private VMs.
- Take regular snapshots for backup and disaster recovery.



BEST PRACTICES

- ✓ Use least privilege IAM roles for VM access.
- ✓ Enable OS Login and disable project-wide SSH keys.
- ✓ Deploy across multiple zones for availability.
- ✓ Use Managed Instance Groups for scaling and reliability.
- ✓ Keep OS and applications updated.
- ✓ Enable monitoring and alerting.
- ✓ Take regular snapshots and test restore.
- ✓ Use VPC firewall rules to restrict traffic.



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GCP Storage



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GCP Storage services provide scalable, durable, secure and cost-effective solutions to store all types of data for your applications.

Google Cloud

1 Cloud Storage



- Object storage for any type of data.
- Highly durable and available.
- Used for backups, media, websites, data lakes, etc.
- Access via HTTPS/JSON API, gsutil, or SDKs.

2 Filestore



- Fully managed NFS file system.
- Lift-and-shift applications that require shared file storage.
- Scalable performance.
- Regional and Enterprise editions.

3 Persistent Disk



- Block storage for Compute Engine VMs.
- Multiple types: Standard, Balanced, SSD, Extreme SSD.
- Persistent even if VM is stopped.
- Zonal (default) and Regional.

4 Local SSD



- High performance local storage attached to VM.
- Data is ephemeral.
- Ideal for caching, scratch data, and high IOPS workloads.

5 Archive Storage



- Long-term, low-cost storage for data rarely accessed.
- Minimum storage duration of 365 days.
- Used for compliance, backups, and cold data.

6 Storage Classes

	Standard	Frequent access (High availability)
	Nearline	Infrequent access (Lower cost)
	Coldline	Rare access (Lowest cost)
	Archive	Archive data (Longest retention)

7 Bucket



- Containers for storing objects in Cloud Storage.
- Buckets must be globally unique.
- Define location, storage class, and access control.

8 Lifecycle Rules



- Automatically manage object lifecycle.
- Move data between storage classes.
- Set object expiration to delete old data.

9 Multi-Regional Storage

- Stores data across multiple regions for durability and availability.
- Options: US (multi-region), EU (multi-region), Asia (multi-region).
- Ideal for global applications and disaster recovery.



10 Summary



Storage Options at a Glance

Service	Type	Durability	Performance	Use Cases
Cloud Storage	Object	11 9's	Standard	Backups, websites, data lakes, media, analytics
Filestore	File (NFS)	99.9%	High	Lift & shift apps, shared file storage
Persistent Disk	Block	99.999%	High	VM boot disk, databases, transactional apps
Local SSD	Block (local)	Data ephemeral	Very High	Caching, scratch data, high IOPS workloads
Archive Storage	Object	11 9's	Low	Long-term archive, compliance, cold data

Key Points

- ✓ Choose the right storage based on access frequency and performance needs.
- ✓ Use lifecycle rules to optimize cost.
- ✓ Multi-regional storage increases availability and durability.
- ✓ Combine services for best performance and cost optimization.



REMEMBER

- Cloud Storage is for objects.
- Filestore is for file systems.
- Persistent Disk is for block storage.
- Local SSD is temporary but very fast.
- Archive Storage is for long-term retention.
- Use lifecycle rules to reduce storage costs.



BEST PRACTICES

- ✓ Use appropriate storage class to control costs.
- ✓ Enable lifecycle rules for automatic data management.
- ✓ Replicate critical data using multi-regional storage.
- ✓ Use CMEK (Customer Managed Encryption Keys) for security.
- ✓ Monitor storage usage and set alerts.
- ✓ Regularly review and delete obsolete data.



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GCP Networking



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Google Cloud's networking services let you build secure, high performance and global scale networks for your applications.

Google Cloud

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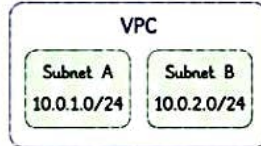
VPC



- Virtual Private Cloud is your isolated network in GCP.
- Create a global VPC network and deploy resources in it.
- Full control over IP ranges, subnets and routes.

2

Subnet



- Subnets are segments of a VPC.
- Regional resources must be in a subnet.
- Define IP ranges for your resources.

3

Firewall Rules



- Stateful firewall for ingress and egress traffic.
- Allow or deny traffic based on rules.
- Targets: All instances, tags or service accounts.

4

Routes



- Routes determine where network traffic is sent.
- Default system routes are created automatically.
- Custom routes for specific destinations.

5

Cloud Router



- High availability router for your VPC.
- Used for dynamic routing (BGP).
- Connects to VPN or Interconnect.

6

Cloud NAT



- Provides outbound internet access to VMs without external IP.
- Uses IP addresses from NAT IP range.
- Improves security & reduces cost.

7

Public IP



- External IP address reachable over the internet.
- Ephemeral (dynamic) or Static (reserved).
- Used for VMs, Load Balancers, NAT, etc.

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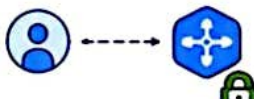
Private Google Access



- Enables VMs in private subnets to access Google APIs & services.
- No public IP or NAT required.
- More secure and efficient.

9

Private Service Connect



- Access Google APIs and partner services privately.
- Uses internal IP addresses.
- Keeps traffic on Google's network.

10

Cloud DNS



- Scalable, reliable and managed DNS service.
- Supports public and private DNS zones.
- High availability with global anycast network.

11

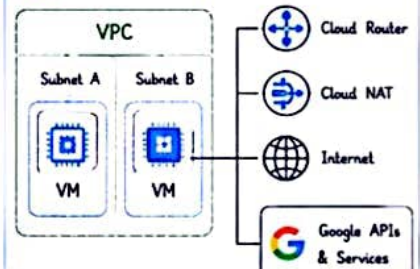
Cloud VPN



- Site-to-site VPN over the internet.
- Secure IPsec encrypted tunnel.
- Connect on-prem to GCP VPC.

12

Networking Overview



Typical Traffic Flow



REMEMBER

- VPC is your isolated network.
- Use custom subnets and firewall rules.
- Use Cloud NAT for secure outbound internet.
- Use Private Google Access for Google services.
- Use Private Service Connect for private access.
- Use Cloud DNS for name resolution.
- Use Cloud VPN/Interconnect for hybrid connectivity.



BEST PRACTICES

- ✓ Follow least privilege in firewall rules.
- ✓ Use subnets across multiple regions for HA.
- ✓ Enable Private Google Access in private subnets.
- ✓ Use Cloud NAT to avoid assigning public IPs to VMs.
- ✓ Use Internal Load Balancer for internal applications.
- ✓ Monitor VPC flow logs and firewall logs.
- ✓ Use labels to organize network resources.
- ✓ Regularly review routes and firewall rules.



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GCP LOAD BALANCING

Secure, scalable and highly available traffic distribution across your applications.



Google Cloud

7

1 Global Load Balancer



- Global anycast IP.
- Distributes traffic across multiple regions.
- Built-in high availability and disaster recovery.
- Works with HTTP(S), TCP, SSL Proxy and UDP.

Best for global applications and multi-region deployments.

2 Regional Load Balancer



- Internal or external regional forwarding.
- Operates within a single region.
- Lower latency and higher performance for local users.
- Supports TCP and UDP.

Best for regional applications with predictable traffic.

3 HTTP(S) Load Balancer



- Layer 7 load balancing for HTTP and HTTPS.
- Content-based routing, URL maps, header-based routing.
- SSL termination and automatic certificate management.

Best for web applications and REST APIs.

4 Internal Load Balancer



- Internal TCP/UDP Load Balancer for private traffic.
- Uses internal IP addresses only.
- Ideal for migrating applications to GCP.
- Works with backends in the same VPC.

Best for internal applications and microservices.

5 Cloud Armor



- Integrated DDoS protection.
- Web Application Firewall (WAF).
- OWASP Top 10 protection.
- Pre-configured and custom security rules.
- IP reputation and geo-blocking.

Protects applications from attacks and malicious traffic.

6 Cloud CDN



- Global content delivery network.
- Caches content at edge locations.
- Reduces latency and improves application performance.
- Works with HTTP(S) Load Balancer.
- Supports dynamic and static content.

Improves performance and reduces backend load.

7 Autoscaler



- Automatically scales backend instances.
- Based on CPU utilization, request count or custom metrics.
- Works with MIG, GKE, and Cloud Run.
- Ensures high availability and optimal cost.

Ensures applications handle traffic dynamically and efficiently.

8 Health Check

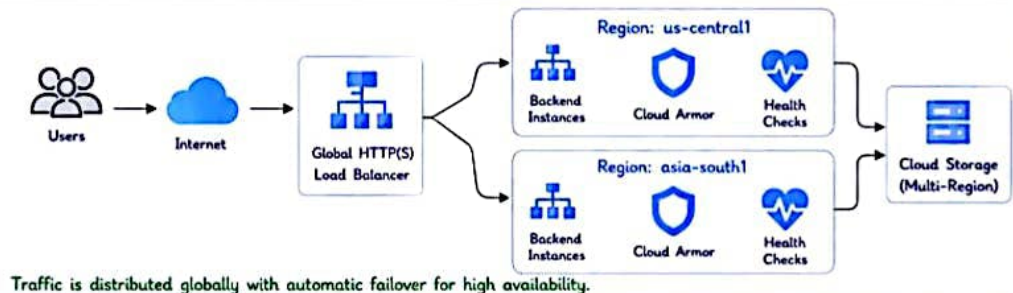


- Monitors backend health.
- Types: HTTP, HTTPS, TCP, SSL, HTTP/2, gRPC.
- Automatic removal of unhealthy backends.
- Configurable check interval, timeout and thresholds.

Ensures traffic is routed only to healthy backends.

9 High Availability

- Global Load Balancer with multi-region backends.
- Health checks and auto-failover.
- Redundant instances across zones and regions.
- Zero single point of failure.
- Works with MIG, GKE and Cloud Run.



10 Best Practices



Use Cloud Armor for security and DDoS protection.



Always configure health checks for your backends.



Enable autoscaling for handling traffic spikes.



Use Cloud CDN to improve performance and reduce latency.



Deploy across multiple regions for high availability.



Use HTTPS Load Balancer with managed SSL certificates.



Monitor with Cloud Monitoring and Log Analytics.



Use consistent naming and labels for easy management.

Load Balancing Types (Quick Reference)

Type	Scope	Layer	Protocol	Use Case
Global HTTP(S) Load Balancer	Global	L7	HTTP/HTTPS	Web apps, APIs, Global apps
Global TCP/SSL Proxy LB	Global	L4	TCP/SSL	Non-HTTP protocols
Regional External LB	Regional	L4	TCP/UDP	Regional apps with static IP
Internal TCP/UDP LB	Regional	L4	TCP/UDP	Internal services in VPC

Cloud DNS

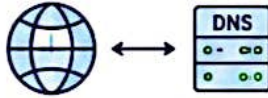


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Cloud DNS is a scalable, reliable and low-latency DNS service to manage your domain name system on Google Cloud.

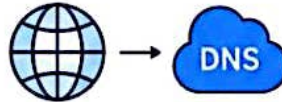
Google Cloud

1 DNS Basics



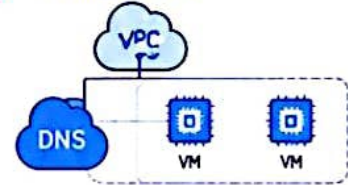
- DNS translates domain names to IP addresses.
- Distributes traffic and improves availability.
- Hierarchical and distributed system.

2 Public Zone



- Hosted on Cloud DNS.
- Resolve domain names on the public internet.
- Example: example.com
- Used for websites, APIs, email, etc.

3 Private Zone

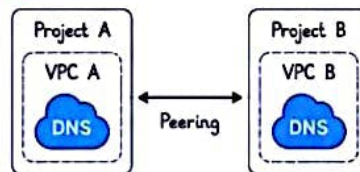


- Resolve names within a VPC network.
- Not accessible from the public internet.
- Example: internal.example.com
- Used for internal apps and services.

4 Record Types

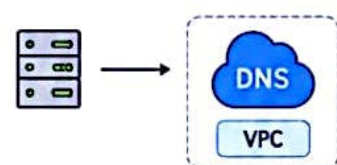
A	Maps domain to IPv4 address
AAAA	Maps domain to IPv6 address
CNAME	Alias of another domain
MX	Mail exchange records
TXT	Text records (SPF, DKIM, etc.)
NS	Name server records
SRV	Service locator record

5 DNS Peering



- Connect networks to share private DNS zones.
- Enables name resolution across VPCs and projects.
- Improves connectivity in hybrid apps.

6 Private DNS



- Use private zones for internal domains.
- Works with VPCs through Network association.
- Keeps internal DNS traffic inside Google network.

7 Resolver



- Inbound: on-premises → Cloud DNS.
- Outbound: Cloud → on-premises DNS.
- Enables hybrid DNS name resolution.

8 Health Check



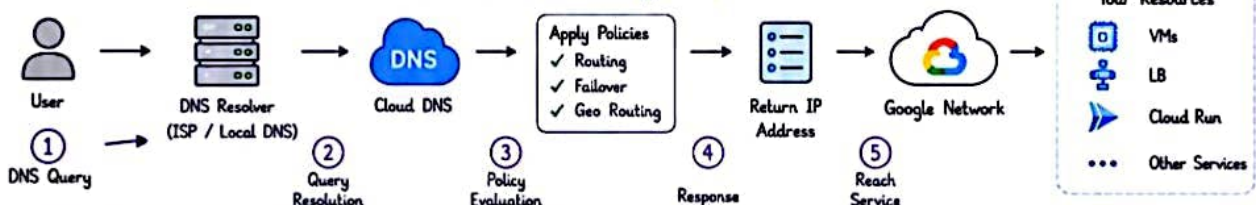
- Monitor your resources (VMs, LB, websites, APIs).
- Used with Cloud DNS for failover routing.
- Supports HTTP, HTTPS, TCP, SSL.

9 Traffic Flow



- User queries DNS for a domain.
- DNS returns best IP based on routing, policy and health.
- Traffic reaches the closest healthy service.

End-to-End DNS Traffic Flow



REMEMBER

- ✓ Cloud DNS is highly available and globally distributed.
- ✓ Use Public Zones for internet-facing domains.
- ✓ Use Private Zones for internal name resolution.
- ✓ DNS Peering allows sharing private DNS across VPCs.
- ✓ Resolver helps connect on-premises and GCP DNS.
- ✓ Health checks improve availability and routing.



BEST PRACTICES

- ✓ Use appropriate DNS record type for your use case.
- ✓ Enable DNSSEC for additional security.
- ✓ Use Geo Routing for global users and low latency.
- ✓ Implement Health Checks with failover policies.
- ✓ Use Private DNS for internal services.
- ✓ Monitor DNS performance and set alerts.
- ✓ Use TTL wisely to balance performance and cache.



App Engine & Cloud Run



9

Fully managed platform to build, deploy and scale applications and APIs on Google Cloud.

Google Cloud

1 App Engine



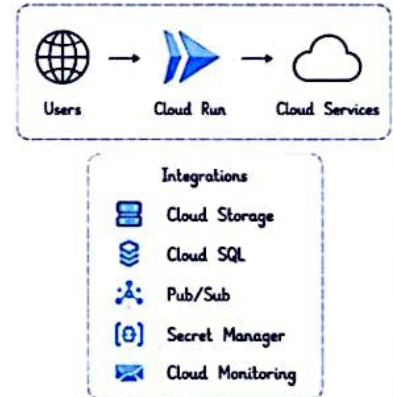
- PaaS service to build and run scalable web applications.
- Supports multiple languages.
- No infrastructure management.
- Automatic scaling and high availability.
- Two environments:
 - Standard (sandboxed)
 - Flexible (Docker based)



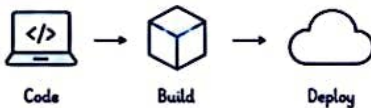
2 Cloud Run



- Run stateless containers in a fully managed environment.
- Built on Knative.
- Scale to zero.
- Pay only for requests and CPU used.
- Ideal for microservices and APIs.



3 Deployment



- Deploy from code, container or artifact.
- App Engine: `gcloud app deploy`
- Cloud Run: `gcloud run deploy`
- Supports CI/CD with Cloud Build, GitHub Actions, Cloud Deploy.

4 Service Configuration



- Configure memory, CPU, concurrency (Cloud Run).
- Set environment variables and secrets.
- Configure VPC access, IAM, and permissions.
- Set timeout, max instances, and CPU allocation.
- Create revisions and traffic splitting.

5 Auto Scaling



- App Engine: automatic scaling based on requests.
- Cloud Run: scales to zero and up to maximum instances.
- Scale based on concurrent requests, CPU utilization and custom metrics.

6 Custom Domain



- Map your domain to App Engine or Cloud Run.
- Add CNAME record to Google managed domain.
- Supports domain mapping and SSL automatically.

7 HTTPS



- HTTPS is enabled by default.
- Google-managed SSL certificates.
- Automatic certificate provisioning and renewal.
- Enforce HTTPS redirect for secure access.

8 Monitoring



- Use Cloud Monitoring for metrics, alerts and dashboards.
- Use Cloud Logging for logs.
- Trace requests with Cloud Trace.
- Error reporting for exceptions.
- Integration with Cloud Profiler.

How App Engine & Cloud Run Work



Common Integrations

- Cloud Storage
- Cloud SQL
- Pub/Sub
- Secret Manager
- Cloud Monitoring

REMEMBER

- App Engine is ideal for full web applications.
- Cloud Run is ideal for microservices and APIs.
- Cloud Run scales to zero - cost effective.
- HTTPS is enabled by default.
- Monitor your applications and set alerts.
- Use managed services for databases and storage.
- Keep services stateless for better scaling.



BEST PRACTICES

- ✓ Choose between App Engine and Cloud Run based on your use case.
- ✓ Keep applications stateless for scalability.
- ✓ Use least privilege IAM for service accounts.
- ✓ Enable HTTP/2 and compression.
- ✓ Use CDN (Cloud CDN) in front if needed.
- ✓ Set proper resource limits (CPU, Memory).
- ✓ Monitor, log and set alerts for errors and performance.
- ✓ Use Cloud Build + Cloud Deploy for CI/CD.



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COST OPTIMIZATION
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DESIGN

OPTIMIZE

SECURE

SCALE

Google Kubernetes Engine (GKE)

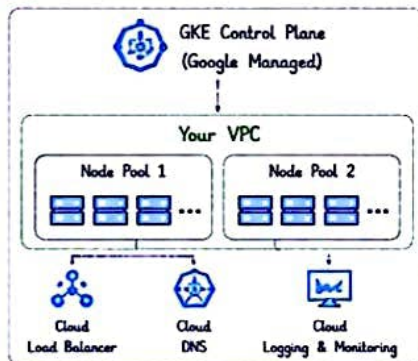


10

GKE is a fully managed Kubernetes service to build, deploy and scale containerized applications on Google Cloud.

Google Cloud

1 GKE Architecture

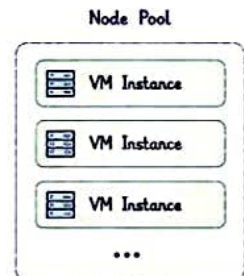


- Control plane managed by Google.
- Worker nodes run in your VPC.
- Highly available, secure and scalable.

2 Node Pool



- A group of Compute Engine VMs that run container workloads.
- Can have multiple node pools with different configurations.
- Supports autoscaling, upgrades and taints/tolerations.



3 Control Plane

- Fully managed by Google.
- Includes API Server, Scheduler, Controller Manager, etc.
- Automatically scaled, upgraded and highly available.
- You never manage the control plane nodes.

4 Pods

- Smallest deployable unit in Kubernetes.
- One or more containers that share network and storage.
- Ephemeral - recreated when needed.
- Managed by Deployments, StatefulSets or Jobs.

5 Services

- Stable endpoint to access a set of Pods.
- Types: ClusterIP, NodePort, LoadBalancer, ExternalName.
- Provides service discovery and load balancing.

6 Ingress

- Manages external HTTP/HTTPS access.
- Uses Ingress Controller (e.g., GCE Ingress).
- Supports host-based and path-based routing.
- Works with TLS/SSL certificates.

7 Horizontal Pod Autoscaler (HPA)



- Automatically scales Pods based on metrics like CPU, memory or custom metrics.
- Keeps your application performant and cost-effective.
- Scale up or down based on demand.

8 Cluster Autoscaler

- Automatically adjusts the number of nodes in node pools.
- Adds nodes when Pods are pending.
- Removes nodes when underutilized.
- Works with HPA for optimal scaling.

9 VPC-native Cluster



- Each Pod gets a unique VPC IP address.
- Better network performance and security.
- Full visibility and control using VPC networking features.
- Recommended for most production workloads.

10 Autopilot Mode

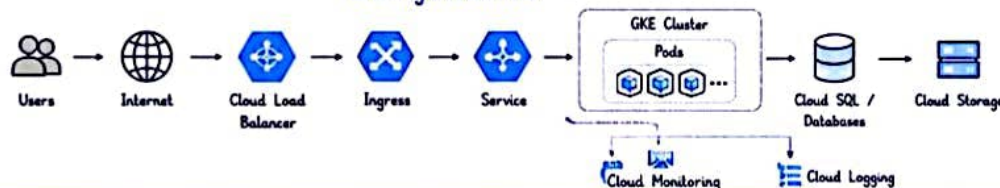


- No node management - Google manages nodes for you.
- Automatic scaling, upgrades and security patching.
- Pay only for Pod resources used.
- Focus only on your applications.

11 GKE Components

API Server	Central management endpoint.
Scheduler	Schedules Pods on nodes.
Controller Manager	Maintains cluster state.
etcd	Consistent key-value store.
kube-proxy	Network proxy on each node.
CoreDNS	Cluster DNS service.

GKE High Level Flow



Common Integrations

- Cloud SQL
- Cloud Storage
- Cloud IAM
- Cloud Monitoring
- Secret Manager

REMEMBER

- GKE - Google Managed Kubernetes.
- Control plane is fully managed by Google.
- Use VPC-native clusters for better networking.
- HPA handles Pod scaling. Cluster Autoscaler handles nodes.
- Autopilot is serverless Kubernetes experience.



BEST PRACTICES

- ✓ Use least privilege IAM for GKE and workloads.
- ✓ Enable binary authorization and workload identity.
- ✓ Use private clusters and authorized networks.
- ✓ Keep nodes and dependencies updated.
- ✓ Enable monitoring, logging and alerts.
- ✓ Use resource requests and limits for Pods.
- ✓ Use Pod disruption budgets for high availability.



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Cloud Build & CI/CD



11

Fully managed CI/CD tooling on Google Cloud to build, test, deploy and release your applications faster and safer.

Google Cloud

1 Cloud Source Repositories



- Fully managed private Git repositories.
- Secure, scalable and highly available.
- Integrates with Cloud Build and Cloud Deploy.
- Supports IAM and fine-grained access control.

2 Cloud Build



- Serverless build service.
- Builds containers, runs tests, creates artifacts.
- Executes builds in isolated environments.
- Scales automatically.
- Supports parallel builds and custom substitutions.

3 Artifact Registry



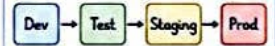
- Fully managed artifact repository.
- Stores Docker images, Maven, npm, Python, Helm charts, etc.
- Secure with IAM and VPC Service Controls.
- Integrated with Cloud Build and GKE.

4 Cloud Deploy



- Continuous delivery service for Kubernetes and serverless.
- Automates release to multiple targets.
- Supports multi-cluster and multi-environment delivery.
- Rollout strategies: canary, blue/green & standard.

5 Cloud Deploy Pipeline



- Define delivery pipeline with stages and targets.
- Each stage promotes the artifact to the next environment.
- Manual approvals and automated checks supported.
- Rollback made easy.

6 Build Triggers



- Automatically start a build on events.
- Trigger on push, pull request or tag.
- Custom filters for branches, files and paths.
- Ensures automated and consistent builds.

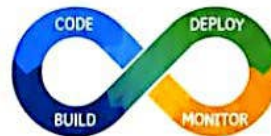
7 YAML Build File



- Define build steps in a cloudbuild.yaml file.
- Specify images, commands, tests and artifacts.
- Reusable, version-controlled and configurable.
- Example:

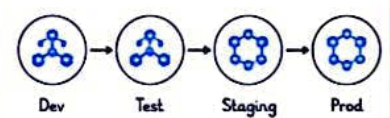
```
steps:
- name: 'gcr.io/cloud-builders/docker'
  args: ['build', '-t', 'gcr.io/$PROJECT_ID/app:1.0', '.']
images:
- 'gcr.io/$PROJECT_ID/app:1.0'
```

8 CI/CD Flow



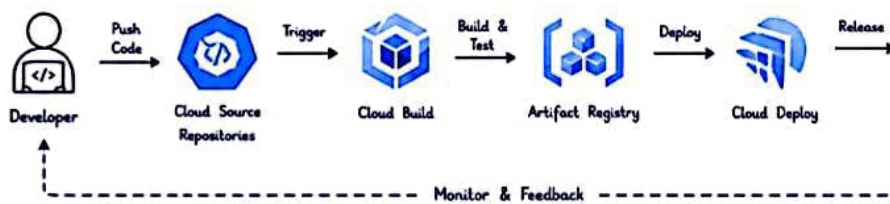
- Commit code -> Trigger build
- Build & test -> Create artifact
- Store artifact -> Deploy to target
- Monitor & get feedback -> Iterate

10 Multi-stage Deployment



- Promote the same artifact across environments.
- Use progressive delivery strategies:
 - Canary
 - Blue/Green
 - Rolling
- Ensures quality, reduces risk and downtime.

End-to-End CI/CD Flow on Google Cloud



Supported Targets

- GKE (Anthos)
- Cloud Run
- VM (Compute Engine)
- Custom Targets

Common Build Steps

Step	Description
Fetch Code	Clone source code
Install	Install dependencies
Build	Compile or build app
Test	Run unit/integration tests
Package	Build container / artifact
Push	Push to Artifact Registry

Cloud Build Substitutions

Variable	Description
\$PROJECT_ID	GCP Project ID
\$BUILD_ID	Unique build ID
\$COMMIT_SHA	Commit SHA
\$BRANCH_NAME	Branch name
\$TAG_NAME	Git tag name
\$SHORT_SHA	Short commit SHA

Cloud Deploy Features

- ✓ Multi-target delivery
- ✓ Automated rollbacks
- ✓ Approval & verification
- ✓ Pipeline templates
- ✓ Scaffold & Helm support
- ✓ History and audit logs
- ✓ Release tracking

Supported Artifact Types

- Docker Images
- Maven Packages
- npm Packages
- Python Packages
- Helm Charts
- Generic Files

Trigger Events

- Push to Branch
- Pull Request
- Tag Creation
- Manual Trigger
- Scheduled Trigger
- Pub/Sub Event

REMEMBER

- ✓ Cloud Build is serverless - no server management.
- ✓ Store all build configs in version control.
- ✓ Use Artifact Registry instead of Container Registry.
- ✓ Automate everything - from build to production.
- ✓ Monitor builds and deployments for reliability.



BEST PRACTICES

- ✓ Use separate projects or environments for isolation.
- ✓ Use least privilege IAM for builds and deployments.
- ✓ Enable vulnerability scanning for container images.
- ✓ Use approval gates for production deployments.
- ✓ Keep build times fast and parallelize steps.
- ✓ Use blue/green or canary for zero-downtime releases.





Gain visibility, ensure reliability and optimize performance of your applications and infrastructure on Google Cloud.

Google Cloud

1 Cloud Monitoring

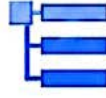


- Collects metrics, events and metadata from GCP resources, applications and services.
- Real-time metrics for infrastructure and applications.
- Integrates with GCP services and custom applications.
- Provides uptime checks and service level monitoring.

Key Use Cases:

Infrastructure monitoring, SLOs, uptime monitoring, custom metrics.

2 Cloud Logging



- Centralized log management service.
- Collects logs from GCP, on-prem, applications and containers.
- Powerful log search with Log Explorer using advanced queries.
- Export logs to BigQuery, Cloud Storage or Pub/Sub.
- Support for audit logs and compliance.

Key Use Cases:

Troubleshooting, audit, security monitoring, operational analytics.

3 Error Reporting

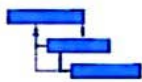


- Automatically detects, groups and reports application errors.
- Supports multiple languages and frameworks.
- Provides stack traces, error context and trend analysis.
- Integrates with Cloud Monitoring and Cloud Logging.

Key Use Cases:

Find, triage and fix application errors faster.

4 Trace



- End-to-end request tracing for distributed applications.
- Visualize latency across services and identify bottlenecks.
- Works with OpenTelemetry and Cloud Trace SDK.
- Correlates with logs and metrics.

Key Use Cases:

Performance troubleshooting, latency analysis, microservices visibility.

5 Profiler



- Continuous CPU and heap profiling for your applications.
- Identify performance issues and memory leaks.
- Supports Java, Go, Python and Node.js.
- Low overhead, always-on profiling.

Key Use Cases:

Performance optimization, CPU usage analysis, memory leak detection.

6 Alerting



- Create alerting policies on metrics, logs and uptime checks.
- Multi-channel notifications: Email, SMS, PagerDuty, Slack, Webhooks.
- Alert on thresholds, absences or anomaly detection.
- Escalation policies and alert documentation.

Key Use Cases:

Incident detection, proactive alerting, on-call notifications.

7 Metrics Explorer



- Interactive tool to explore metrics.
- Build queries, visualize charts and compare metrics.
- Diagnose performance issues and capacity planning.
- Save charts and share insights.

Key Use Cases:

Metric analysis, trend analysis, capacity planning, root cause analysis.

8 Dashboard



- Create custom dashboards with rich visualizations.
- Combine metrics, logs, traces and uptime checks.
- Share dashboards with your team.
- Real-time insights in one place.

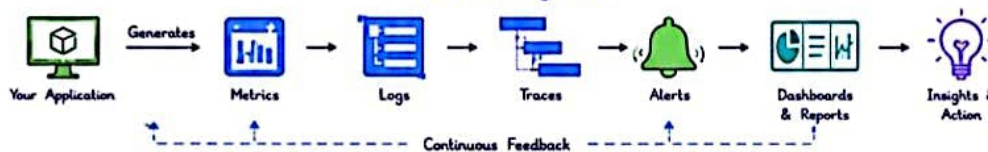
Key Use Cases:

Operational dashboards, NOC view, executive reporting.

9 Other Integrations

	Cloud Security Command Center	Security posture & threat monitoring
	Cloud Asset Inventory	Asset visibility and inventory
	Service Level Objectives (SLOs)	Reliability tracking and error budgets
	Uptime Checks	Monitor availability from external locations
	Synthetic Monitoring	Test user workflows and availability

Observability Flow



Data Sources

- GCP Services
- VMs & Instances
- Kubernetes (GKE)
- Applications
- Third-party Tools

REMEMBER

- Monitoring = Metrics (what happened) + Logs (why it happened) + Traces (how it happened).
- Set meaningful alerts to reduce noise.
- Correlate metrics, logs and traces for faster root cause analysis.
- Define SLOs and track error budgets.
- Review dashboards regularly.



BEST PRACTICES

- Use consistent naming and labeling for all resources.
- Monitor RED/USE metrics for services.
- Create dashboards per environment and team.
- Use log-based metrics for custom alerting.
- Set alert thresholds based on baseline and trends.
- Automate incident response with alerting integrations.
- Regularly review SLOs and error budgets.



Secure your applications, data and infrastructure
across Google Cloud.

Google Cloud

① Security Command Center



Unified security management and risk visibility.

- Centralized view of security posture.
- Detects threats and vulnerabilities.
- Asset inventory and risk scoring.
- Compliance monitoring and findings.
- Integrates with Cloud services and third-party tools.

Key Benefit: Better visibility, earlier detection and faster response.

② Cloud Armor



- DDoS protection and WAF.
- Layer 3, 4 and 7 protection.
- Preconfigured and custom rules.
- Geo-based access control.
- Bot management and threat intelligence.

Key Benefit: Protects applications from attacks and abuse.

③ Secret Manager



Securely store, manage and access secrets.

- Store API keys, DB passwords, tokens.
- Automatic rotation of secrets.
- Fine-grained IAM access control.
- Audit logs for all secret access.
- Integrates with GKE, Cloud Run, VMs and more.

Key Benefit: Keep secrets safe and reduce risk of leakage.

④ Cloud KMS



Manage encryption keys with ease and control.

- Customer-managed encryption keys.
- Symmetric and asymmetric keys.
- Key rotation and versioning.
- Integrated with Cloud services.
- Hardware security backed by Google.

Key Benefit: Protect data with strong encryption and key management.

⑤ IAM (Identity & Access Management)



Control who can do what on your resources.

- Manage users, groups and service accounts.
- Grant least-privilege access.
- Roles: Basic, Predefined and Custom.
- IAM Conditions for fine-grained access.
- Audit logs for all IAM changes.

Key Benefit: Enforce least privilege and secure access.

⑥ Firewall Rules



Control traffic at the VPC network level.

- Allow or deny traffic by IP, port, protocol.
- Ingress and egress rules.
- Target tags, service accounts or IP ranges.
- Priority based rule evaluation.
- Logs integrated with Cloud Logging.

Key Benefit: Protect network resources with controlled access.

⑦ VPC Service Controls



Prevent data exfiltration across GCP services.

- Define security perimeters around services.
- Restrict access to resources.
- Prevent unauthorized data leaving perimeters.
- Works with IAM for fine control.
- Supports dry-run mode for testing.

Key Benefit: Protect sensitive data from internal and external threats.

⑧ Cloud IDS



Threat detection for network traffic.

- Detects malicious activity in VPC traffic.
- Powered by Suricata IDS engine.
- Supports inline and passive modes.
- Works at subnet or firewall level.
- Integrates with Security Command Center.

Key Benefit: Detect and stop threats early in the network.

⑨ DDoS Protection

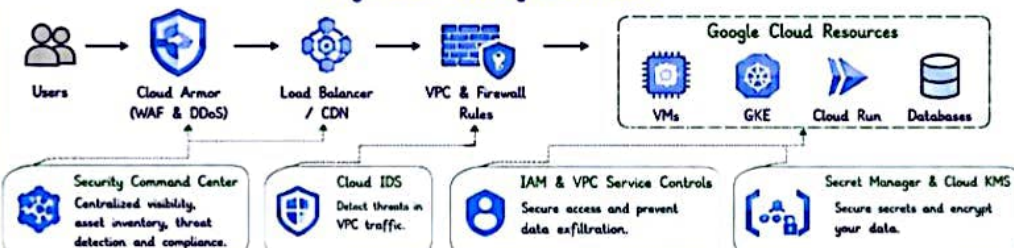


Always-on protection against DDoS attacks.

- Automatic and always enabled.
- Network layer (L3/L4) protection.
- No configuration required.
- Global anycast infrastructure.
- Works with Cloud Armor for L7 protection.

Key Benefit: Keep your applications available and resilient.

Google Cloud Security Architecture



Integrated Services

- Cloud Logging
- Cloud Monitoring
- Cloud Audit Logs
- Chronicle Security
- BigQuery (Security Analytics)

REMEMBER

- ✓ Follow the principle of Least Privilege.
- ✓ Enable MFA for all users.
- ✓ Regularly rotate keys and secrets.
- ✓ Monitor and review logs and alerts.
- ✓ Keep your systems and dependencies updated.
- ✓ Use defense-in-depth strategy.



BEST PRACTICES

- ✓ Use Security Command Center for continuous monitoring.
- ✓ Enable Cloud Armor for all public-facing applications.
- ✓ Store secrets in Secret Manager, not in code.
- ✓ Encrypt data using Customer Managed Keys (Cloud KMS).
- ✓ Use IAM Conditions and least-privilege roles.
- ✓ Apply VPC Service Controls for sensitive workloads.
- ✓ Enable Cloud IDS for threat detection.
- ✓ Review firewall rules regularly.



Backup & Disaster Recovery



14

Protect your data, ensure business continuity and recover faster with Google Cloud.

Google Cloud

1 Backup and DR Service



- Fully managed backup and recovery service.
- Centralized management for GCP resources.
- Policy based backups with scheduling and retention.
- Supports backup for VMs, databases and file systems.
- Cross-region disaster recovery orchestration.

★ Benefit: Simplifies backup management and accelerates recovery.

2 Snapshots



- Point-in-time, incremental, read-only copies of Persistent Disks.
- Stored in the same or different region.
- Fast, space-efficient and cost-effective.
- Can be used to create new disks or restore.
- Automatic snapshots with schedules.

★ Benefit: Quick recovery and easy data protection.

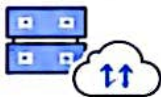
3 Persistent Disk Backup



- Back up Persistent Disks using Backup and DR Service.
- Application-consistent backups with quiesce support.
- Define backup vault, policy, schedule and retention.
- Restore to original disk or create a new disk.
- Supports cross-region backup and restore.

★ Benefit: Reliable VM and data volume protection.

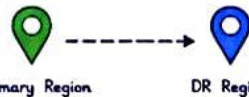
4 Cloud Storage Backup



- Versioning to keep multiple versions of objects.
- Lifecycle rules to transition or archive data.
- Object retention and retention lock (WORM).
- Regular exports or sync to other buckets/regions.

★ Benefit: Durable, low-cost and scalable object data protection.

5 Cross-Region Replication



- Replicate data across different regions for disaster recovery.
- Supported for Cloud Storage, Cloud SQL, Spanner, Bigtable and more.
- Reduces risk of data loss due to regional outages.
- Asynchronous replication for performance and cost efficiency.

★ Benefit: Higher availability and protection against regional failures.

6 Multi-region Buckets



- Store data across multiple regions automatically.
- High availability and durability.
- Built-in redundancy and resilience.
- Ideal for global applications and compliance needs.
- Supported locations: US, EU, ASIA.

★ Benefit: Global data availability with built-in resilience.

7 Recovery Planning



- Identify critical systems and RTO/RPO.
- Design backup strategy and replication plan.
- Document failover and fallback procedures.
- Test recovery regularly.
- Monitor, review and update DR plan.

★ Benefit: Preparedness reduces downtime and recovery time.

8 Best Practices



- Follow 3-2-1 rule: 3 copies, 2 media, 1 offsite.
- Automate backups and validate regularly.
- Use appropriate retention based on business needs.
- Test restore to ensure backups are usable.
- Monitor backup jobs and set alerts.
- Use least privilege IAM for backup operations.

★ Benefit: Strong backup posture and faster business continuity.

RPO vs RTO

RPO (Recovery Point Objective)



Maximum acceptable data loss measured in time.

Example: RPO = 15 mins (you can lose max 15 mins of data)

RTO (Recovery Time Objective)



Maximum acceptable time to restore the service.

Example: RTO = 1 hour (service restored within 1 hour)

3-2-1 Backup Rule



3 Copies of your data

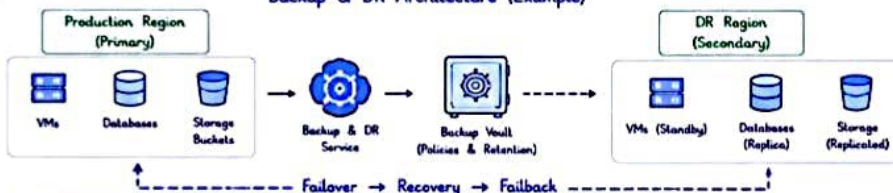


2 Different media types



1 Copy offsite or off-cloud

Backup & DR Architecture (Example)



KEY TAKEAWAYS

- Backups protect your data.
- Replication protects your applications.
- Plan and test your recovery.
- Monitor and improve continuously.



BEST PRACTICES SUMMARY

- Automate and validate backups.
- Use cross-region replication for critical data.
- Test DR plan regularly.
- Monitor backup and replication health.
- Keep documentation up to date.



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SECURE

SCALE

GCP PRODUCTION ARCHITECTURE



15

Complete Production Architecture

Highly available, secure and scalable production architecture with monitoring, security, backup and disaster recovery.

Google Cloud

